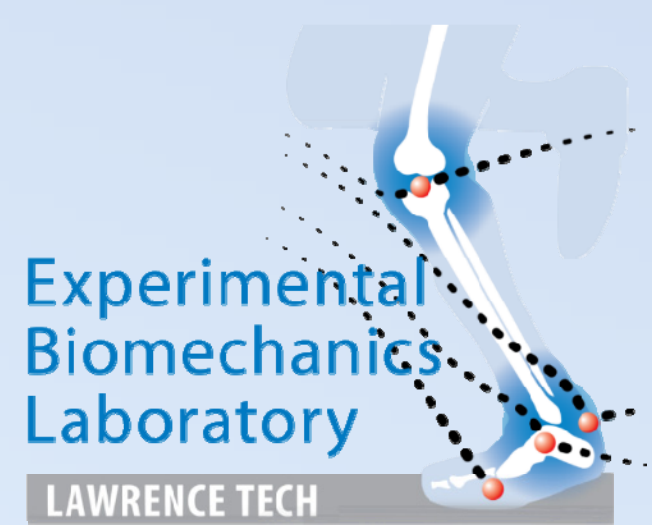




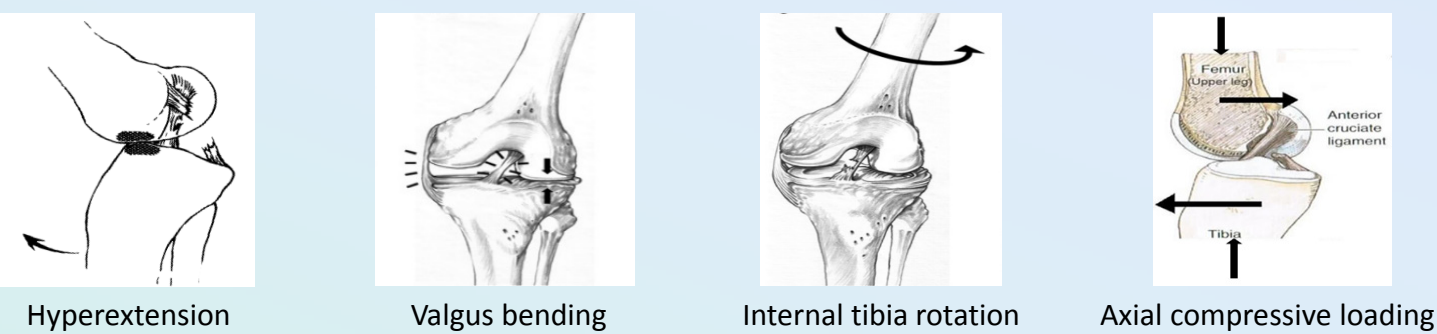
Novel Design of an Anterior Cruciate Ligament (ACL) Injury Prevention Brace

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Background

- In the United States, 200,000 - 300,000 ACL injuries occur annually.
- Common ACL injury mechanisms include: hyperextension, valgus (inward) bending, internal rotation of the tibia, and axial compressive loading of the knee.
- Although some athletes wear knee braces to reduce the risk of ACL injury, those braces only protect against hyperextension and valgus bending.
- Another type of knee brace commonly worn is an osteoarthritis (OA) brace. OA braces function to reduce the compressive load on the knee's medial compartment.

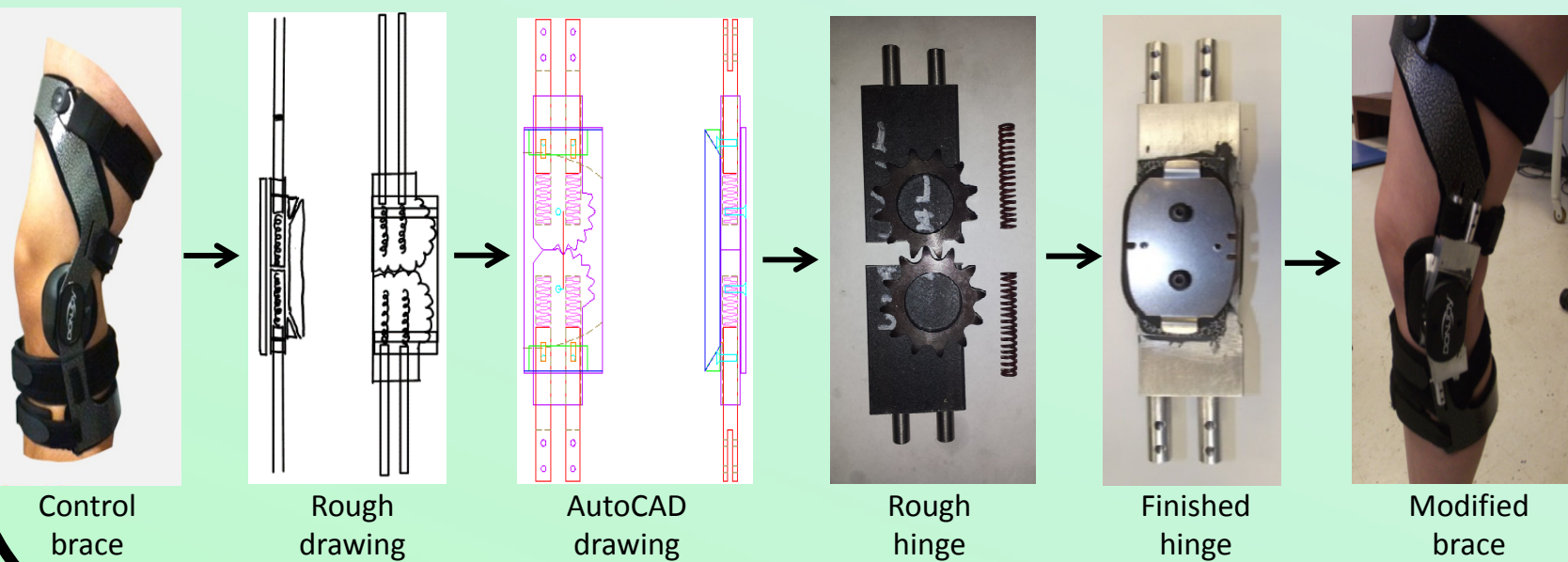


Project Goal: To redesign the medial hinge mechanism of a prophylactic knee brace used to prevent ACL injuries by shifting the axial compressive knee reaction force from the lateral to medial compartment of the knee. The hinge will serve to reduce the occurrence of ACL injuries by protecting against hyperextension, valgus bending, and axial compressive loading.

Conceptual Hinge Design

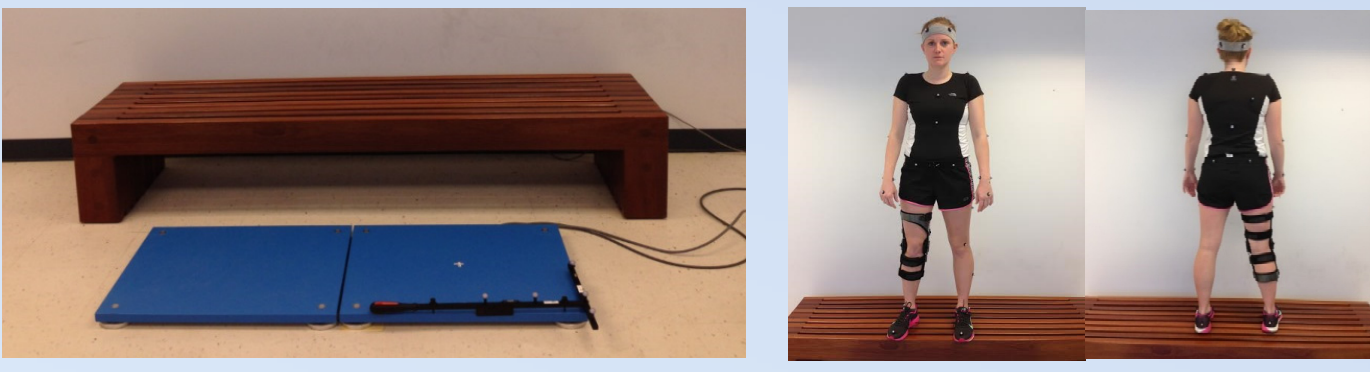
- The hinge continues to protect against or reduce the severity of ACL injuries due to hyperextension and valgus bending while allowing knee range of motion during athletic movements.
- It orients the knee to have a slightly varus bending angle compared to baseline anatomy.
- Part of the vGRF are converted into a lateral brace displacement when the knee is near full extension.

These specifications should create a varus bending angle of the knee during jump landing or rapid deceleration movements which would stabilize the knee joint by partially unloading compression across the lateral compartment, therefore reducing the risk to the ACL.



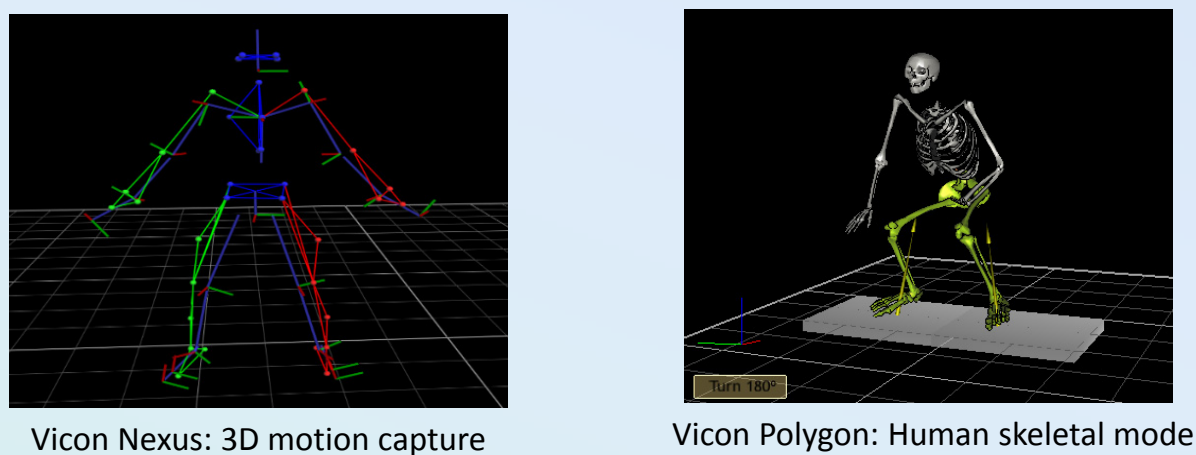
Testing

Experiment

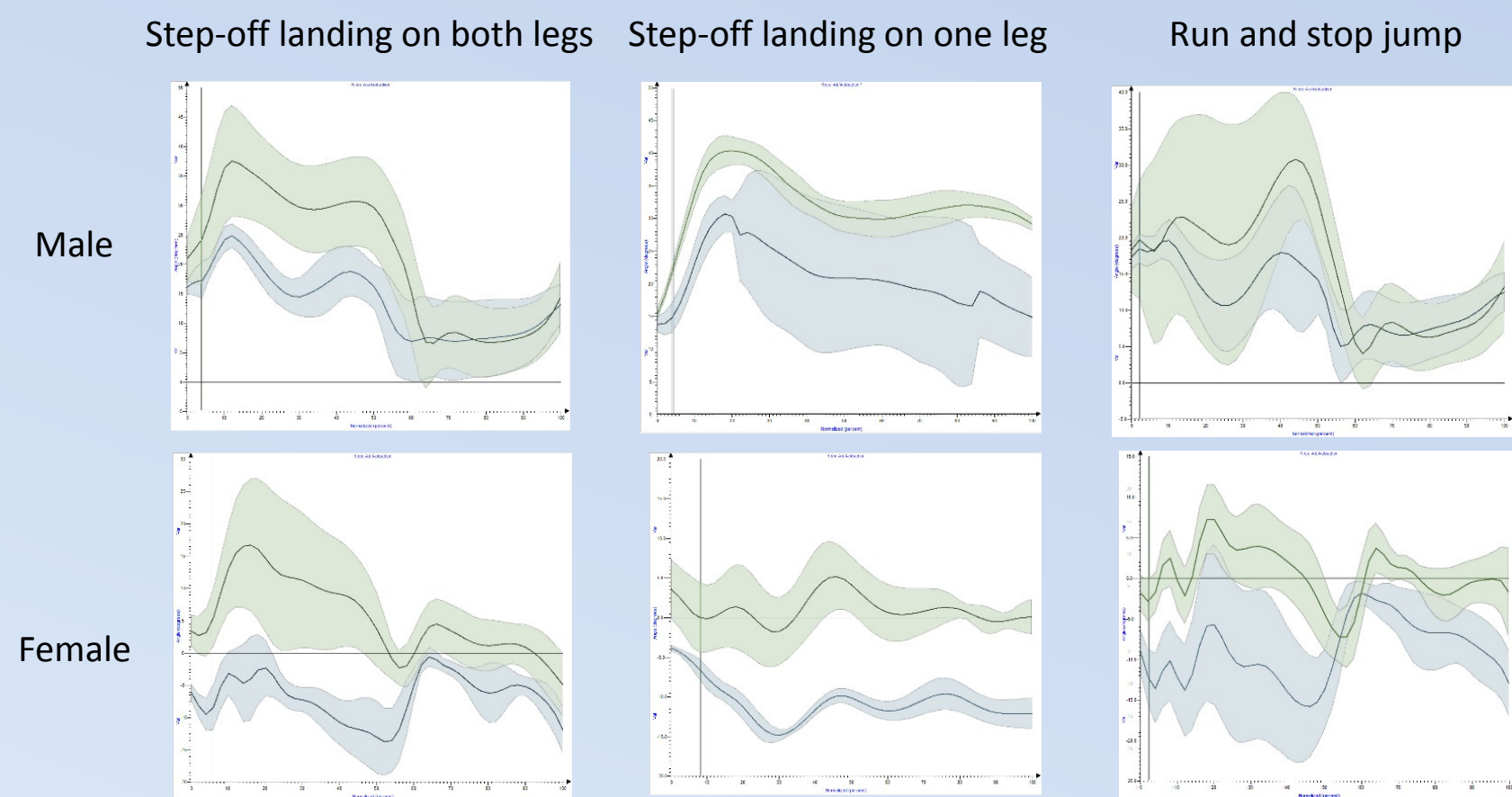


- 3 experimental tests were completed for both the control and modified braces.
 - Step-off landing on both legs
 - Step-off landing on one leg
 - Run and stop jump

Data Analysis



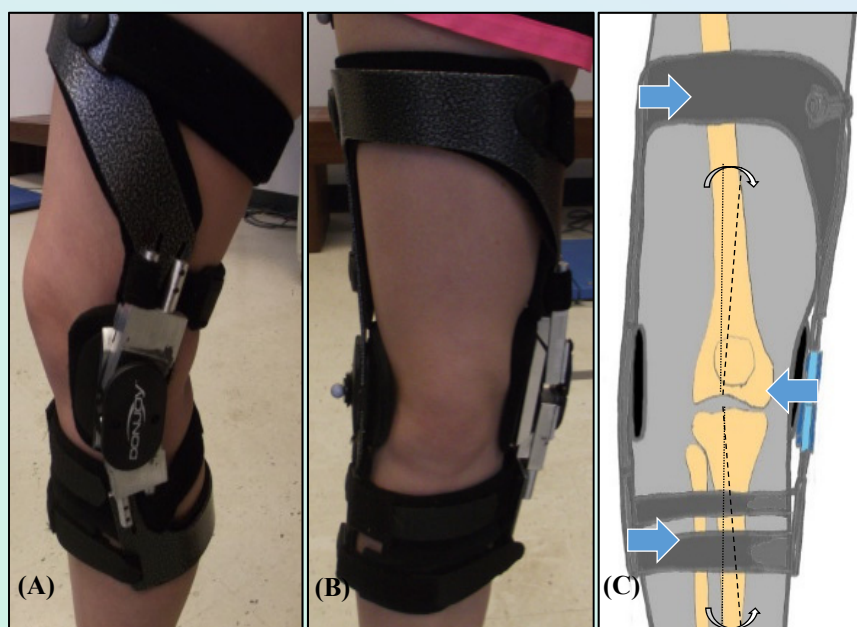
Results



- Average and standard deviations during five repeated trials. Blue: control brace, Green: modified brace
- Varus angle increased by 17° for male participant and 28° for female participant during the step-off landing on both legs tests with the modified brace.
 - Varus angle increased by 8° for male participant and 10° for the female participant during the step-off landing on one leg tests with the modified brace.
 - Varus angle increased by 10° for male participant and 15° for the female participant during the run stop-jump landing with the modified brace.

Design Outcome

- Replace medial hinge of Donjoy's Armour knee brace.
- Lateral hinge was not modified.
- Modified hinge is held to frame by four rods.
- Rods are connected to two wedges.
- Wedges move in conjunction with the rods, serving to compress springs when a load is applied.
- The two wedges move central to the knee joint.
- Wedges press against a wedge pad to create a lateral brace movement at the knee joint.
- Creates a varus bending moment applied to the knee.



Modified brace hinge design; (A) medial view, (B) front view, (C) slight varus angle induced due to lateral displacement of the medial brace hinge.

Conclusion/Future Work

- The modified brace shows an increase in the knee varus angle when compared to the control brace.
- A decreased knee moment value is observed while using the modified brace.
- The compressive load on the lateral side of the knee was reduced.
- Future applications will further analyze the experimental tests as well as continue to develop the functioning mechanisms of the hinge.

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